

EXPT NO:2	LOAD AND VISUALIZE A DATASET USING TABLEAU and PowerBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. Why is Tableau a preferred tool for data visualization?

Tableau is widely used because it makes data easy to understand through interactive and visually appealing dashboards. It allows users to create charts and reports without needing advanced programming knowledge. The drag-and-drop interface helps in building visualizations quickly, even for beginners. Tableau can connect to different types of data sources such as Excel, SQL databases, and cloud platforms. It also supports real-time data analysis, making business decisions faster. These features make Tableau an effective Business Intelligence tool for analyzing and presenting data.

2. What is the primary use of a line chart in data analysis?

A line chart is mainly used to show how data changes over a period of time. It helps identify trends, growth, declines, or recurring patterns in continuous data. Businesses often use line charts to monitor sales, revenue, stock prices, or customer activity over days, months, or years. Since the data points are connected by lines, it becomes easier to compare changes between different time periods. This visualization supports better forecasting and performance tracking. As a result, line charts are one of the most common tools in Business Intelligence reporting.

3. When would you use a tree map instead of a bar chart?

A tree map is useful when there are many categories and limited screen space. Instead of displaying bars, it represents data using rectangles of different sizes, where the area reflects the value of each category. This makes it easier to understand the proportion of each item within the whole dataset. Tree maps are commonly used for sales by product category, market share, or budget allocation. They provide a quick overview of the largest and smallest contributors. Compared to a bar chart, a tree map is more effective for showing part-to-whole relationships.

4. What is the significance of filters in Tableau visualizations?

Filters are important because they allow users to display only the information that is relevant to their analysis. Instead of viewing the complete dataset, users can focus on specific regions, products, time periods, or customer groups. This makes dashboards more interactive and easier to explore. Filters also help reduce unnecessary information, making patterns and insights clearer. In Business Intelligence, they support faster decision-making by allowing users to analyze data from different perspectives. Overall, filters improve both the usability and effectiveness of Tableau dashboards.

5. How does Tableau handle geographic data for visualizations?

Tableau has built-in support for geographic data, making it simple to create maps without complex setup. It automatically recognizes fields such as country, state, city, postal code, and latitude or longitude. Once the location data is identified, Tableau plots it on interactive maps. Users can also add colors, sizes, or labels to represent business measures like sales, profit, or population. Geographic visualizations help identify regional trends and compare performance across different locations. This feature is especially useful in Business Intelligence for location-based analysis and strategic planning.

IN-LAB EXERCISE:

OBJECTIVE:

To load a dataset into Tableau, create a variety of visualizations, and derive insights from the data using different chart types available in Tableau.

PROCEDURE:

Dataset Selection:

Use the Global Superstore Sales Dataset (available on Kaggle or Tableau's sample datasets).

Ensure the dataset includes fields like Order Date, Category, Region, Sales, Profit, and Customer Segment.

Data Loading:

Open Tableau and connect to the dataset.

Review the dataset structure, field data types, and missing values.

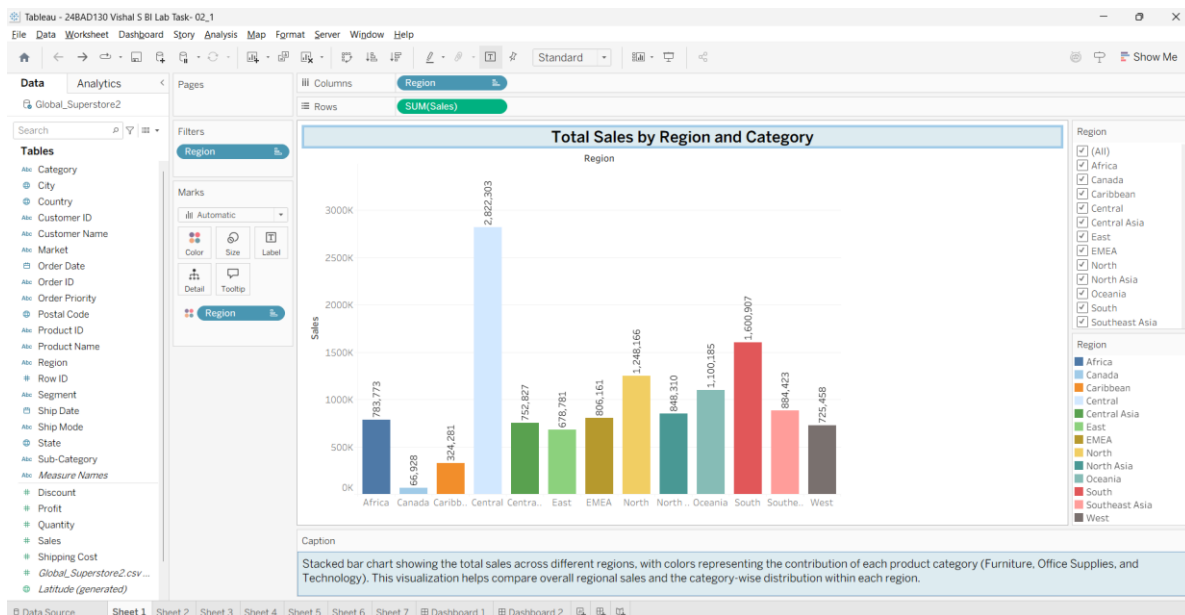
Chart Creations (Step-by-Step):

a) Bar Chart:

Objective: Compare total sales by region and category.

Plot Region on the x-axis and Sales on the y-axis.

Add Category as a color dimension.

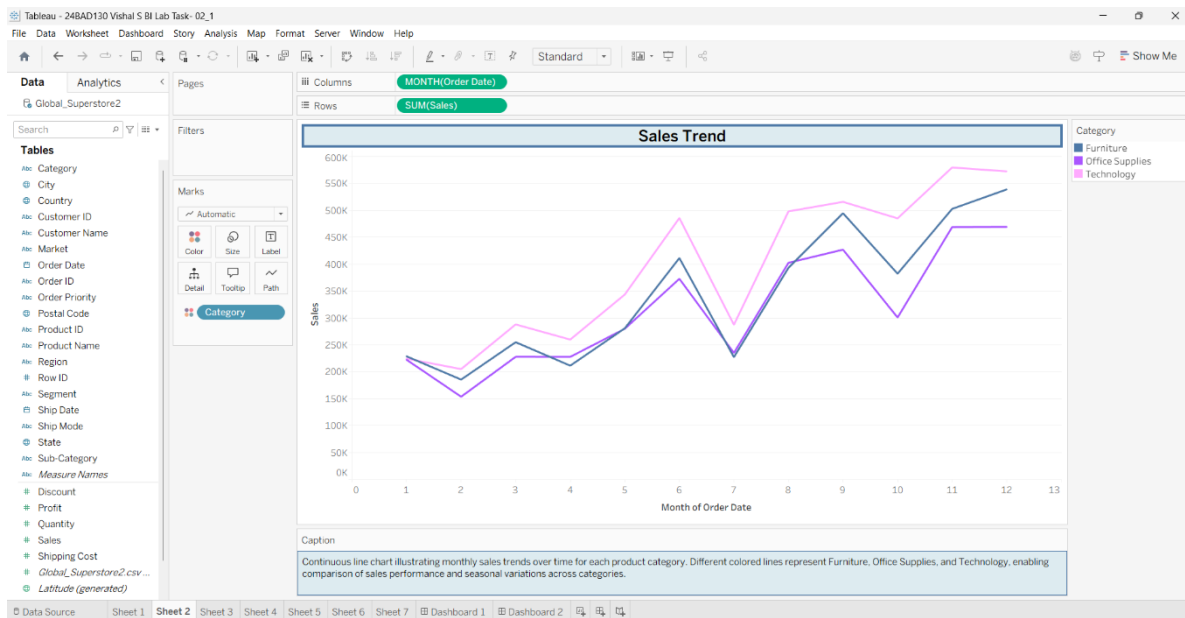


b) Line Chart:

Objective: Visualize sales trends over time.

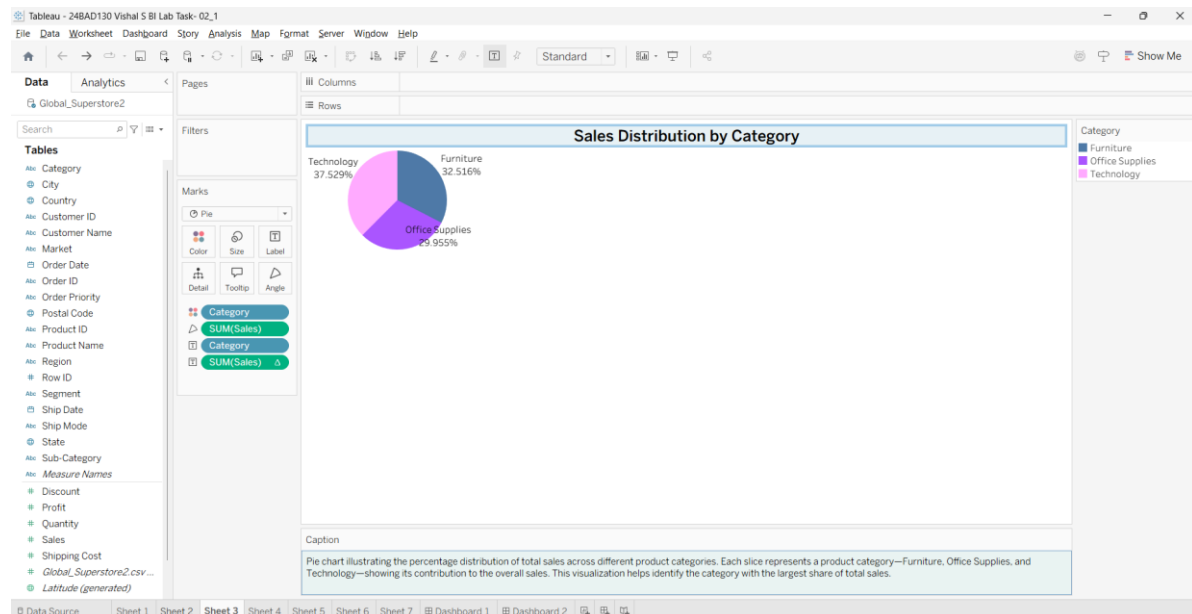
Place Order Date on the x-axis and Sales on the y-axis.

Set granularity to Month and color by Category.



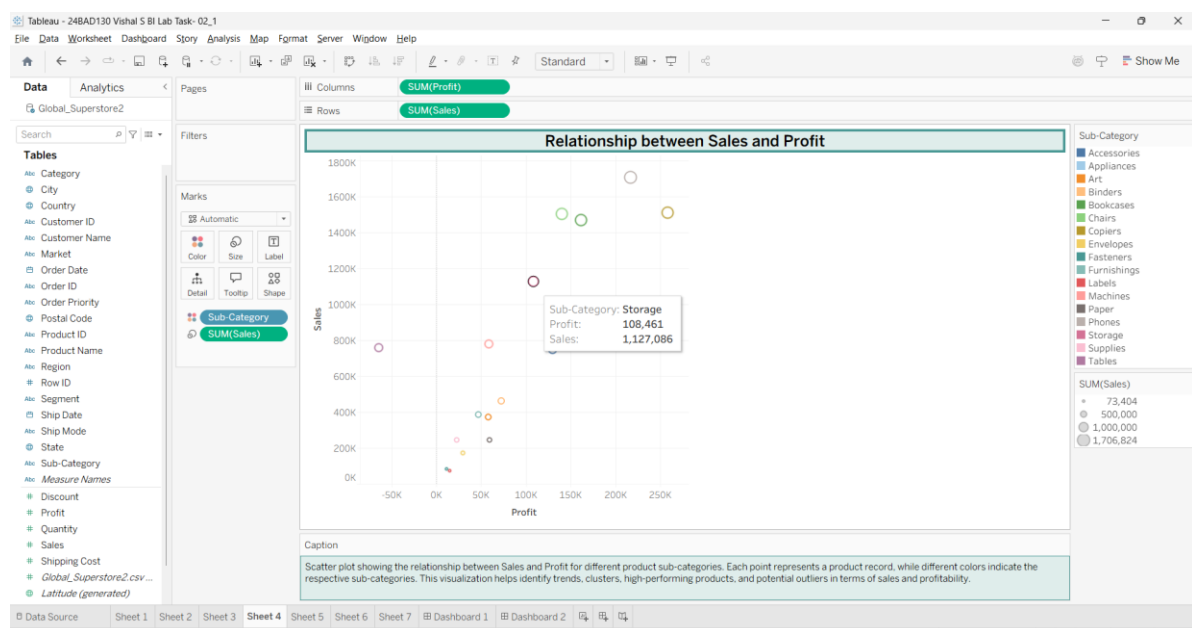
c) Pie Chart:

Objective: Show sales distribution by product category. Assign Category as the dimension and Sales as the measure.



d) Scatter Plot:

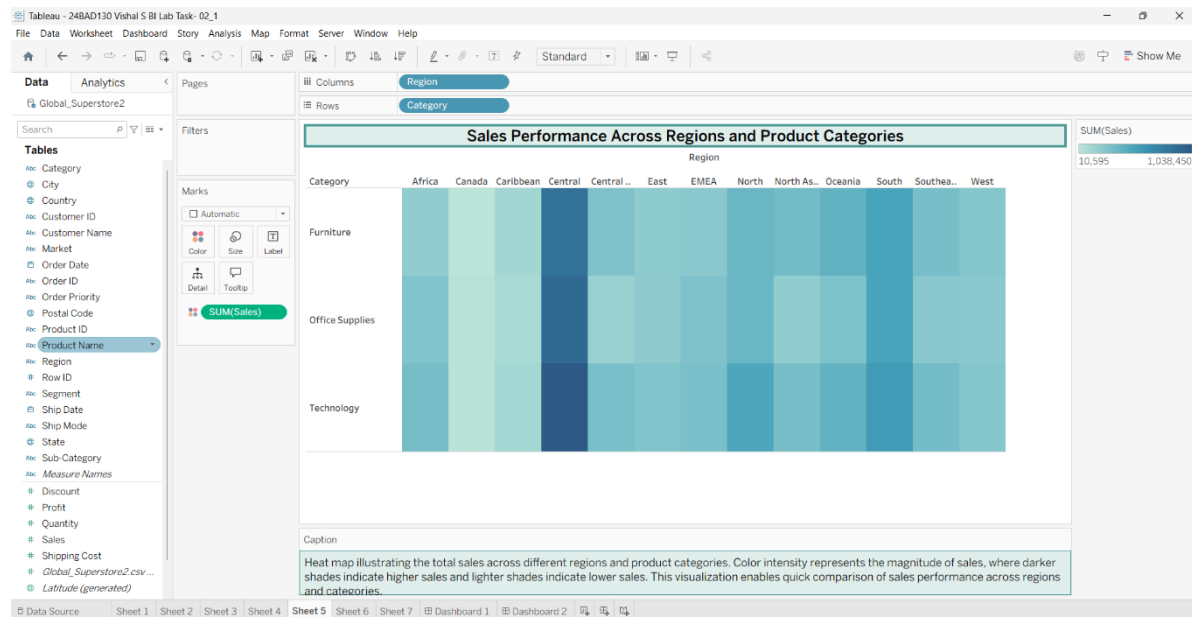
Objective: Examine the relationship between profit and sales.
Plot Profit on the x-axis and Sales on the y-axis.
Use Sub-Category as the color dimension.



e) Heatmap:

Objective: Compare sales across categories and regions.

Use Region and Category as dimensions. Use Sales as the measure and apply color coding.

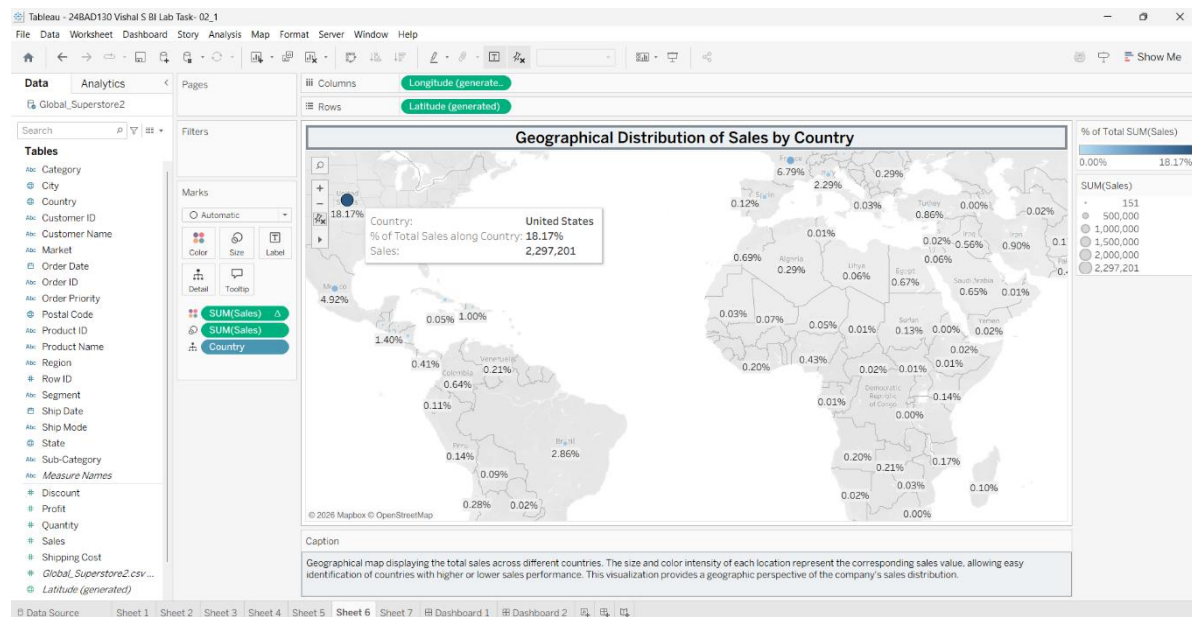


f) Geographical Map:

Objective: Visualize total sales by country.

Set Country as the geographical field. Use

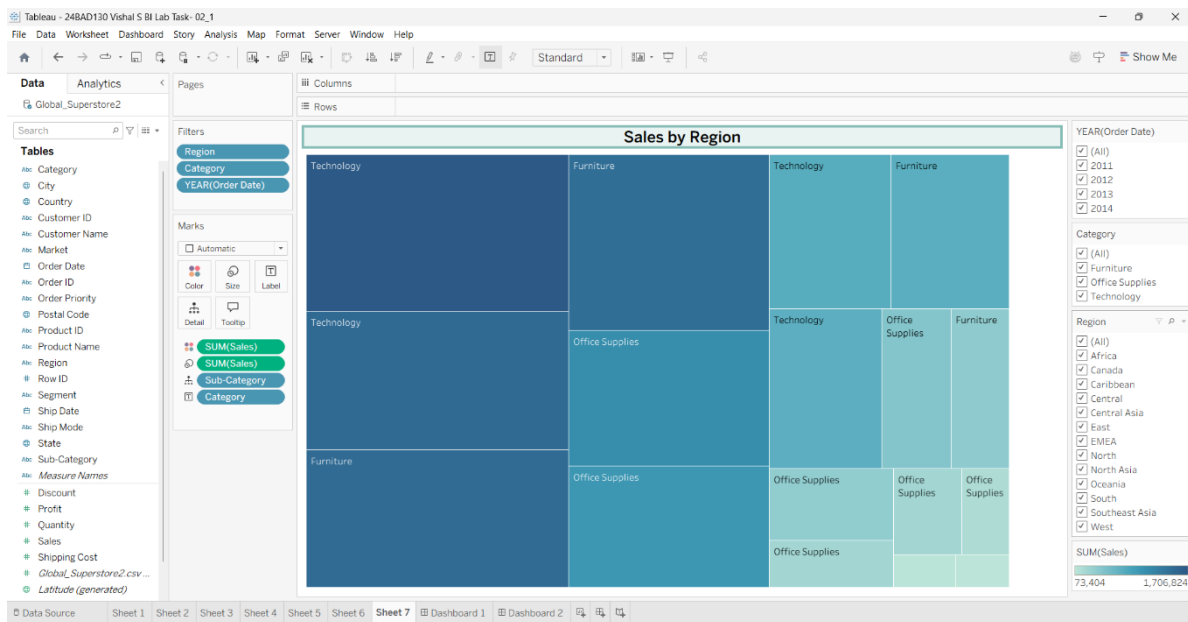
Sales to define size and color gradients.



g) Tree Map:

Objective: Display sales distribution by category and sub-category.

Use Category and Sub-Category as dimensions, and Sales as the measure.



Customization and Interactivity:

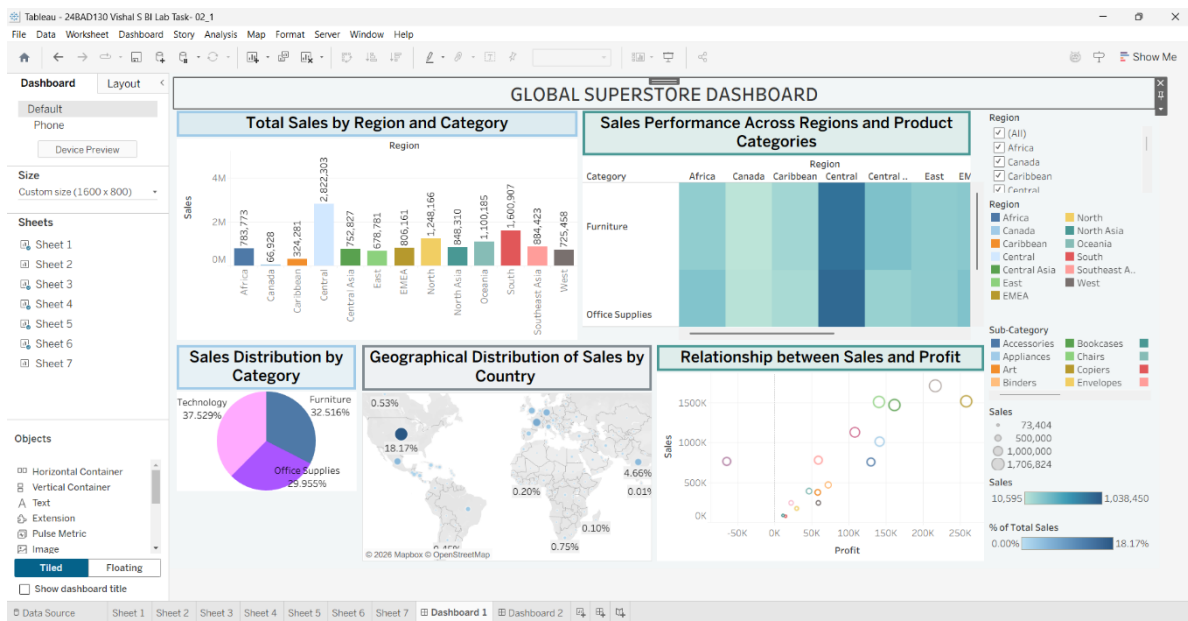
Add filters to explore data subsets interactively (e.g., by year, region, or category).

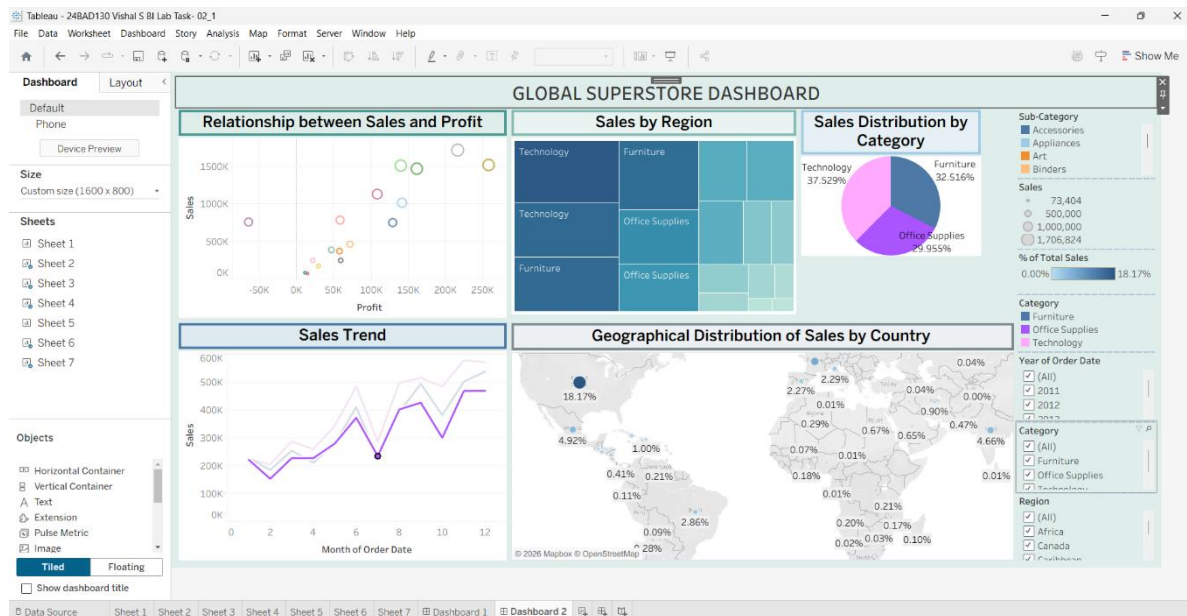
Include annotations, labels, and formatting for clarity.

Dashboard Creation:

Combine all charts into a single dashboard to present a cohesive story.

Add interactive filters and tooltips for dynamic exploration.





Experiment Dataset:

Global Superstore Sales Dataset

Source: <https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset>

Fields Used:

Order Date

Region

Category

Sub-Category

Sales

Profit

Country

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

Bar Chart:

1. What regions had the highest and lowest total sales?

From the bar chart, the Central region recorded the highest total sales, with sales exceeding 2.8 million, making it the best-performing region. The Caribbean region showed the lowest sales, with only around 324 thousand, indicating comparatively weaker business performance. Other regions such as South and North also performed well but remained below Central. The chart clearly highlights the variation in regional sales. This comparison helps businesses identify high-performing and low-performing markets. Such insights are useful for planning future sales strategies and resource allocation.

2. How does the performance of categories vary across regions?

The heatmap shows that category performance is not the same across every region. **Technology** generally performs strongly in most regions, while **Furniture** and **Office Supplies** show moderate sales depending on the location. Regions such as **Central** and **South** have consistently higher sales across all categories. On the other hand, smaller regions contribute lower sales regardless of category. This variation indicates that customer demand differs from one region to another. Businesses can use these insights to optimize product availability based on regional preferences.

Line Chart:

1. Identify the month with peak sales.

The line chart indicates that sales gradually increased throughout the year, with the highest sales occurring in Month 11 (November). Although sales remained high during December, November reached the overall peak. A noticeable decline is seen around Month 7, followed by a strong recovery in the later months. This pattern suggests increased customer purchases during the end-of-year season. Businesses can use this trend for inventory planning and promotional campaigns. Understanding monthly sales patterns supports better forecasting and decision-making.

2. Which category shows consistent growth in sales over time?

Among the three categories, Technology demonstrates the most consistent growth throughout the year. Its sales increase steadily with only minor fluctuations compared to the other categories. Furniture and Office Supplies also show growth but experience more noticeable ups and downs. The Technology category maintains strong performance, especially during the final months of the year. This indicates stable customer demand and better revenue generation. Such information helps businesses prioritize high-performing product categories.

Pie Chart:

1. Which category contributes the most to overall sales?

According to the pie chart, Technology contributes the largest share of total sales at approximately 37.53%. This makes it the leading product category in terms of revenue generation. Furniture follows with about 32.52%, while Office Supplies contributes around 29.96%. Although all categories perform well, Technology clearly has the highest market contribution. This insight helps businesses focus more on products with stronger demand. It also supports better investment and inventory decisions.

2. Are there any categories with negligible sales contributions?

No category has a negligible contribution to total sales. The pie chart shows that all three categories contribute a significant portion of overall revenue. Even Office Supplies, which has the smallest share, contributes nearly 30% of total sales. This indicates that customer demand is well distributed across all product categories. Therefore, each category plays an important role in the company's business performance. Businesses should continue supporting all categories while focusing more on top-performing ones.

Scatter Plot:

1. Is there a noticeable correlation between profit and sales?

Yes, the scatter plot shows a positive relationship between profit and sales. As sales increase, profit generally increases as well, indicating a positive correlation. Most data points are concentrated in the upper-right area, representing high sales with higher profits. However, a few points generate moderate sales but relatively lower profits, suggesting differences in product margins. Overall, higher sales usually lead to better profitability. This relationship helps businesses identify profitable products and sales opportunities.

2. Which sub-category shows the highest sales but low profit?

The scatter plot suggests that some sub-categories generate high sales without producing equally high profits. Although the dashboard legend does not clearly identify the exact sub-category, this pattern indicates products with lower profit margins despite strong sales performance. Such products may involve higher operating costs or frequent discounts. Identifying these items helps businesses review pricing and cost management strategies. Improving the profitability of these products can increase overall business performance.

Heatmap:

1. Which region-category combination has the highest sales?

The heatmap indicates that the Central region combined with the Technology category has the highest sales. This cell appears with the darkest color, representing

stronger sales performance compared to other combinations. Other combinations in the South and North regions also perform well but remain slightly lower. The visualization makes it easy to compare multiple categories across different regions. These insights help businesses focus on the most profitable markets. Regional product planning can be improved using this information.

2. Identify any regions with poor sales performance across all categories.

The heatmap shows that Caribbean and Canada have comparatively lower sales across most product categories. Their lighter color shades indicate weaker business performance when compared to larger regions such as Central and South. These regions may require additional marketing efforts or product improvements. Businesses can investigate local customer preferences to improve sales. Such analysis helps in identifying underperforming markets. It also supports better regional business planning.

Geographical Map:

1. Which country has the highest total sales?

The geographical map clearly shows that the United States has the highest total sales, contributing approximately 18.17% of total global sales. The largest and darkest map marker is located in the USA, indicating its strong market performance. Other countries contribute much smaller percentages in comparison. This suggests that the United States is the company's most significant revenue-generating market. Businesses should continue strengthening their presence in this region. Geographic analysis supports effective market expansion strategies.

2. Are there regions with minimal or no sales?

Yes, the geographical map displays several countries with very small sales contributions, many below **1%** of total sales. These areas are represented by smaller and lighter-colored markers. Although sales exist in these regions, their contribution is minimal compared to major markets like the United States. This indicates opportunities for business expansion and targeted marketing campaigns. Businesses can study these regions to understand customer demand. Such analysis supports future market growth.

Tree Map:

1. Which sub-category dominates the sales in each category?

The tree map shows that larger rectangles represent sub-categories with higher sales within each product category. The biggest blocks are mainly associated with Technology products, indicating that they dominate sales. Furniture and Office Supplies also contain strong-performing sub-categories but occupy comparatively smaller areas. The tree map provides a quick comparison of contribution within each category. Businesses can easily identify which products generate the highest revenue.

This helps in inventory planning and product prioritization.

2. Are there any sub-categories with unexpectedly low sales?

Yes, the tree map contains several very small rectangles that represent sub-categories with relatively low sales. These products contribute only a small portion of the overall category revenue. Their lower performance may be due to limited customer demand or lower market popularity. Businesses should analyze these products further before making inventory decisions. Some may require promotional strategies, while others may need to be replaced. Such analysis improves product portfolio management.

Dashboard:

1. What insights can you derive when combining all visualizations?

The dashboard provides a complete overview of business performance by combining multiple visualizations. It shows that **Technology** is the highest-selling category, the **Central region** performs the best, and the **United States** contributes the largest share of global sales. The line chart highlights seasonal sales growth, while the scatter plot confirms a positive relationship between sales and profit. The heatmap and tree map help identify strong and weak regional performances. Together, these visualizations support better business decisions. The dashboard makes complex data easier to understand and analyze.

2. How do filters enhance the utility of the dashboard?

Filters make the dashboard more interactive by allowing users to view specific data based on region, category, year, or sub-category. Instead of analyzing the entire dataset, users can focus on the information that is most relevant to their needs. This saves time and improves the accuracy of business analysis. Filters also make it easier to compare different segments and identify important trends. Decision-makers can quickly explore multiple scenarios without modifying the original data. As a result, filters greatly improve the usefulness and flexibility of a Business Intelligence dashboard.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		